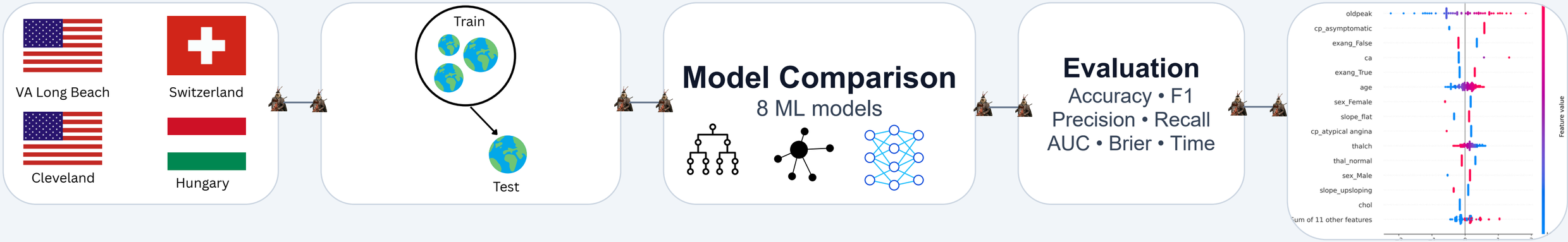
Cross-Source Generalization and Explainability of Machine Learning Models in Heart Disease Classification

Unseen-country evaluation + computational efficiency + country-specific SHAP analysis



Main Findings

SVM
strongest classification
Accuracy 0.777
F1-score 0.809
Recall 0.855

Logistic Regression
efficient
AUC 0.795
Inference time 0.0002 s
Lowest inference time

CatBoost
practical balance
AUC 0.789
Inference time 0.0007 s
Good performance-efficiency balance

SHAP findings:

cp_asymptomatic • oldpeak • chol
Important across datasets, but feature importance and direction can vary by country